

Week 6: Thoughts on Chapter 14 – Evaluation Studies

It seems that an important benefit of mobile usability testing equipment apart from avoid the costs of a usability lab is really that the testing can be done in the setting of use. The chapter mentions this briefly by saying that “another advantage is that equipment can be taken into work settings, enabling testing to be done on site, making it less artificial and more convenient for the participants.” I think the mention of less artificiality in particular is an interesting point, but the chapter does not expound any further on this topic. I believe the reason why testing done on the site where users will actually be using the product in the real world is important, is that it provides external validity which testing everyone at the usability lab will not provide.

External validity is important because it demonstrates that the results found through the usability testing are likely to be what users encounter in their real-life setting. Even if researchers make an effort to make the lab look like what something in the real world would be, at the very best it can only be a rough approximation, and for a fact cannot possibly replicate certain factors that would only exist in a real-world environment. For example, the feeling of having many people bustling busily in your vicinity at an office combined with your being aware of the relevance of their issues to your own work is not something that can ever be replicated externally of your own office environment. External validity is provided by testing users with the mobile equipment because you will be testing their performance on a task in the exact setting where they may actually have to do this task.

The chapter mentions that field studies are really the type of studies meant to provide external validity by testing in the natural setting with little to no control. And it mentions that in these cases usually only qualitative accounts and descriptions are considered. However, I think that it's possible to combine the basic definition of field study with some kind of quantitative-measure experimental design through something which I learned has been defined as *quasi-experimental design* in the field of HCI (from the book *Ways of Knowing in HCI*). Quasi-experiments employ some interesting experimental design types called *non-equivalent groups design* and *interrupted time-series design*. To mention firstly, I found these design methods to be very creative and interesting, but one drawback is that they may be fairly complex and sometimes very costly to arrange, since it is basically researchers doing their best to facilitate experimental design elements in a real-life setting.

Non-equivalent groups design aims to measure changes in performance that result from some intervention. In this case, the intervention would usually pertain to some change in an interface feature that we want to study. In interrupted time-series design, the intervention is applied to different groups at different times across a set period of time in order to control for effects that may result from a specific point in time, such as a political event.